

Solution Requirements

Date	02 October 2026
Team ID	6aba2e98ca93a2750963fe65
Project Name	PocketSmart AI - Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can register with name, email and password, then log in through session or token-based authentication.	High
2	Authorization levels	Authenticated users can access the dashboard, Home/Party/Jewelry planners, recommendation history and protected recommendation details.	High
3	External interfaces	Uses Gemini AI for recommendations and connects through product/search links for Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO-style sources.	High

4	Transactions processing	Validates planner inputs, sends Home/Party/Jewelry requests to the AI service, returns recommendations, and stores request/response history for logged-in users.	High
5	Reporting	Provides a user dashboard, recommendation history and session data so users can review recent and previous planning results.	Medium
6	Business rules, etc.	Budget must be greater than zero; party guests must be 1-10000; jewelry images must be image files within the upload limit; recommendations are budget-aware and exact prices must be verified.	High
7	Compliance to laws or regulations	Protect account/session data and uploaded images; use uploaded outfit images only for broad color/style analysis and do not identify the person.	High
8	Other	Provides three core planners - Home Interior, Party and Jewelry - with optional outfit image input for Jewelry and fallback plans when the AI service is unavailable.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Planner forms and validation should respond promptly; AI and fallback paths should return a usable result without unnecessary retries.	Prototype target: typical planner response within 3-5 seconds when AI service is available.
2	Scalability	The FastAPI routes, services and models are modular so more users, planner categories and recommendation sources can be added without redesigning the whole system.	Prototype target: support at least 100 concurrent sessions in a basic load test.
3	Security & Data Privacy	Passwords are stored as hashes; session/JWT authentication protects user data; uploaded files are type-checked and size-limited before processing.	Acceptance: no plain-text passwords; protected endpoints reject unauthenticated access; non-image uploads are rejected.
4	Reliability & Availability	Home, Party and Jewelry fallback functions provide basic budget plans when the Gemini service is unavailable; health/startup endpoints expose service status.	Acceptance: fallback result is returned when AI calls fail; health endpoint reports status=ok.
5	Usability & Accessibility	Simple forms collect budget, room/event/jewelry details and present structured recommendations for the three planner modules.	Acceptance: all three planners are reachable from the UI, inputs are clearly labeled, and validation errors are understandable.
6	Other	Maintainable modular structure separates routes, models, authentication, catalog logic, AI generation, templates and database access.	Acceptance: planner-specific changes can be made within their module without changing unrelated modules.